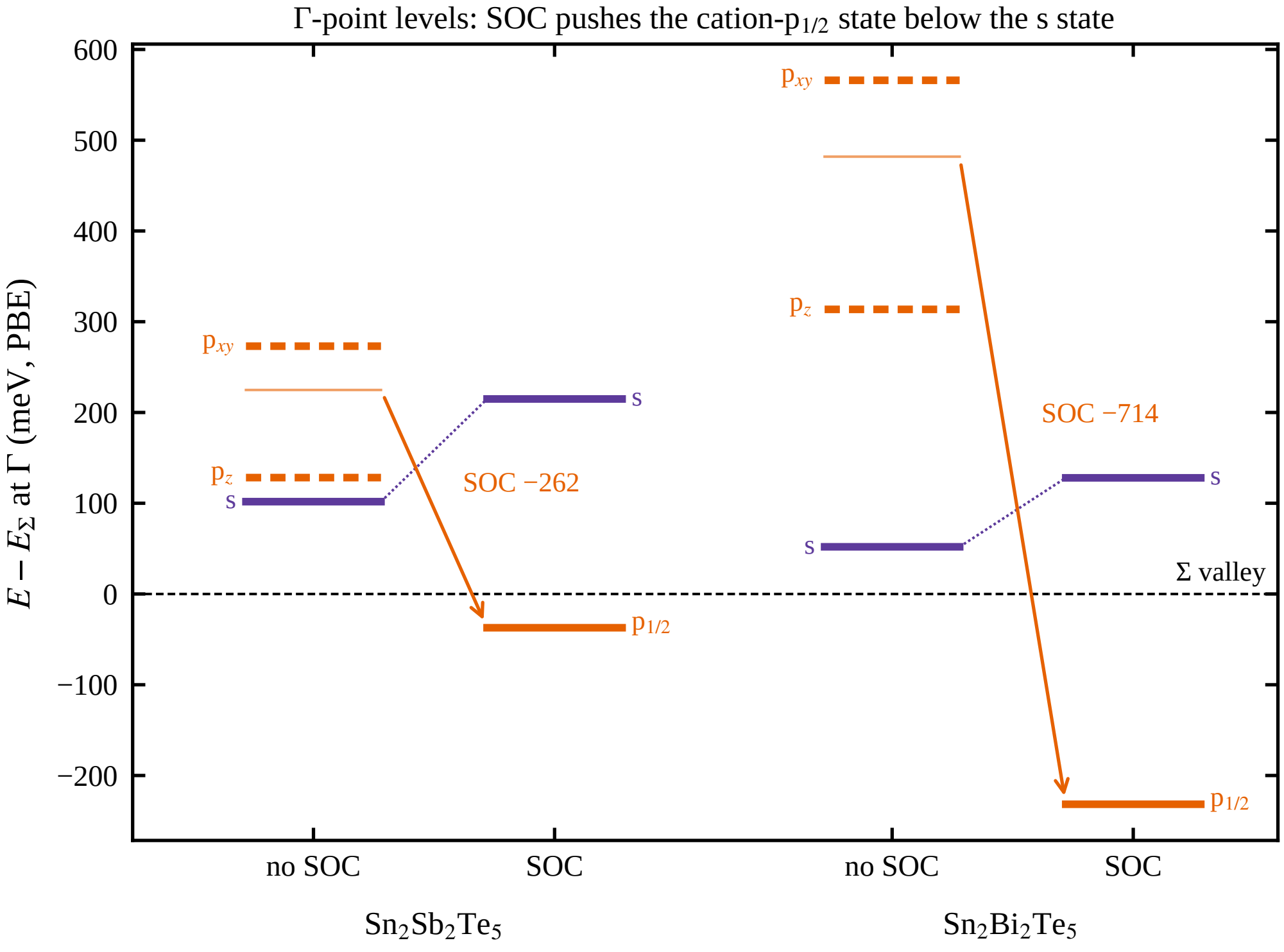
Γ-point levels: SOC pushes the cation-p1/2 state below the s state
E − EΣ at Γ (meV, PBE)
600
500
400
300
200
100
0
−100
−200
pxy
pz
s
SOC −262
p1/2
pxy
pz
s
SOC −714
s
p1/2
Σ valley
no SOC
SOC
no SOC
SOC
Sn2Sb2Te5
Sn2Bi2Te5